

Reproducibility Report: Flores and Coppock (2018)

Do Bilinguals Respond More Favorably to Candidate Advertisements in English or
in Spanish?

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This repository holds the actively maintained replication code for Flores and Coppock (2018), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi [10.1017/S1049096521000366](https://doi.org/10.1017/S1049096521000366)) to a set of published archives.

Article	10.1080/10584609.2018.1426663
Replication archive	10.7910/DVN/XGQ2ZA
Pre-analysis plans	osf.io/e6zjk

The data are not redistributed here. The deposit is 3.7 MB across 14 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See LICENSE.

To reproduce. Clone or download the repository, open `flores_coppock_2018.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 14 files, produces every table and figure into `maintained/output/`, and rebuilds the ground truth from what it wrote. Required packages: `tidyverse`, `estimatr`, `marginalEffects`, `knitr`, `kableExtra`, `here`. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about ten seconds once the deposit is on disk. A successful run overwrites `maintained/output/`, which is committed: **git diff on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. All eight R scripts execute without error on R 4.6.0, with no hardcoded paths beyond a commented instruction to set the working directory and no calls that have since been removed. Every package they name still installs from CRAN, and all of them are alive: `margins` 0.3.28 was released in July 2024, `ggstance` 0.3.7 in April 2024, `coefplot` 1.2.9 in September 2025. The archive’s own README omits `margins` from its list of dependencies, which is the only thing standing between a fresh session and a clean run, and it installs in one line.

Running is also reproducing. The ground truth holds 550 rows, 503 of which pair a published value with a value a deposited script prints. Every one of those 503 matches the article to the precision it reports. There are no exceptions: not one coefficient, standard error, sample size, log likelihood or AIC in the article or its appendix disagrees with what the deposited code produces eight years later.

Two things are nonetheless wrong with the deposit, and neither shows up as an error.

Running the archive in place destroys part of it. Both figure scripts end in `ggsave(filename = "flores_coppock_figure_1.pdf", ...)` with a bare relative path, and both of those PDFs are themselves deposited files. A reader who does what the scripts instruct, setting the working directory to the archive folder and sourcing them, overwrites two of the fourteen deposited files and adds a stray `Rplots.pdf`. Nothing about the result changes, since the re-rendered figures carry the same estimates; what changes is that the deposit no longer verifies against its own published checksums.

The README undercounts the archive. It lists seven scripts and describes the archive as containing them. There are eight. `FC_bilinguals_simulation.R`, which draws Figure 3, appears nowhere in it.

1.2 Does the maintained rewrite reproduce the paper?

Yes, without exception. All 541 claims with a published value to compare against match it to the precision the article reports, and none fails. That includes the 17 values of Table 2 that no deposited script computes, which the rewrite recovers from the deposited `study_1.csv`.

The rewrite contains exactly two numbers not computed from data, 0.05 and -0.15 in `figure_3_simulation.R`. They are not estimates. They are the supposition the article states on page 18 in order to draw Figure 3: “Suppose for the moment that the positive effect among bilinguals is 5 percentage points but the effect among English-only monolinguals is negative 15 percentage points.” Every other value in `maintained/output/` traces to the deposited data.

2 Paper overview

Citation: Flores, A. and Coppock, A. (2018). “Do bilinguals respond more favorably to candidate advertisements in English or in Spanish?” *Political Communication*, 35(4), 612-633. DOI: 10.1080/10584609.2018.1426663

Summary: Three survey experiments ask whether a candidate gains by speaking Spanish to bilingual voters, and what it costs when the same appeal reaches people who do not speak Spanish. Each uses a real advertisement produced by the campaign in matched English and Spanish versions, so the language of the appeal varies while its content does not: Jeb Bush in the 2016 Republican presidential primary, Filemon Vela running for Congress in Texas, and Mike Coffman running for Congress in Colorado. Experiments 1 and 2 add a second randomized factor, the language of the survey itself, giving a 2 by 2 design among bilinguals; Experiment 3 fields the Vela and Coffman ads to English-speaking monolinguals, where only the ad language varies. Among bilinguals the Spanish ad raises general election support for Bush and for Vela by about 5 percentage points and does nothing for Coffman. Among monolinguals the sign flips and the magnitude grows: support for Coffman falls 18.7 points. Taking the survey in Spanish raises bilinguals’ reported sense of linked fate with other Latinos.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Produces	Status on current R
FC_bilinguals_main_analysis.R	Tables 5 to 9	Clean
FC_bilinguals_in_text_stats.R	Table 3	Clean
FC_bilinguals_interaction_analysis.R	Tables A.1 to A.4	Clean
FC_bilinguals_pid_analysis.R	Tables B.5 to B.8	Clean
FC_bilinguals_logit_analysis.R	Tables C.9 and C.10	Clean once margins is installed; the README does not name it
FC_bilinguals_figure_1.R	Figure 1	Clean; <code>geom_errorbarh()</code> warns that it is deprecated; overwrites a deposited file
FC_bilinguals_figure_2.R	Figure 2	Clean; same deprecation warning; overwrites a deposited file
FC_bilinguals_simulation.R	Figure 3	Clean; absent from the README

Every package the archive names installs from CRAN and every one of them has been updated within the last two years: `stargazer` 5.2.3, `estimatr` 1.0.6, `coefplot` 1.2.9, `ggstance` 0.3.7, `margins` 0.3.28, `tidyverse`, `scales`. The only calls that draw a warning are `geom_errorbarh()` in the two figure scripts, deprecated in `ggplot2` 4.0.0 in favour of `geom_errorbar(orientation = "y")`, and `ggstance::position_dodgev()`, which warns about overlapping intervals as it always has. Both still draw the figure correctly.

3.1 The archive overwrites itself

The two figure scripts finish with

```
ggsave(filename = "flores_coppock_figure_1.pdf", g, width = 7, height = 5)
```

and `flores_coppock_figure_1.pdf` is one of the fourteen files in the deposit. Running the archive as its own comments instruct therefore replaces two deposited files with fresh renderings and leaves a `Rplots.pdf` behind. Verifying the four affected checksums is the only way to notice, since the output is correct and no error is raised.

Table 2: What a single in-place run of the archive changes.

File	MD5 as deposited	After one run in place
<code>flores_coppock_figure_1.pdf</code>	<code>f382d6ec39e490d6bf47f9259781ea37</code>	differs, and differs again on every run
<code>flores_coppock_figure_2.pdf</code>	<code>a4de3cc99fbe1147ef702f18e21024fe</code>	differs, and differs again on every run
<code>Rplots.pdf</code>	not in the deposit	created

No fixed checksum can be quoted for the overwritten files, because a PDF records the time it was written and so hashes differently on every run. What can be quoted is how far the new rendering is from the old one. The re-rendered figures are not pixel-identical either, since `ggplot2`'s text metrics have moved since 2017; rasterized at 100 dpi, mean absolute pixel intensity differs by 0.9 percent for Figure 1 and 2.4 percent for Figure 2. The difference is confined to glyph rendering and the small shift in panel geometry that follows from it. The plotted estimates are the same. This is damage to the archive's verifiability, not to its result.

`download_original.R` restores the deposit from Dataverse and checks all fourteen files, so the damage is recoverable for anyone who notices it. The lesson generalizes past this archive: run a deposit's scripts in a copy, never in the deposit.

4 Number-by-number comparison

Table 3: Published values by table, against the deposited scripts and against the maintained rewrite.

Location	Claims	In archive output	Archive matches	Archive fails	Rewrite matches	Rewrite fails
<code>table_1</code>	1	0	0	0	0	0
<code>table_2</code>	17	0	0	0	17	0
<code>table_3</code>	19	19	19	0	19	0
<code>table_4</code>	1	0	0	0	0	0
<code>table_5</code>	31	31	31	0	31	0
<code>table_6</code>	31	31	31	0	31	0
<code>table_7</code>	31	31	31	0	31	0

table_8	31	31	31	0	31	0
table_9	10	10	10	0	10	0
figure_1	1	0	0	0	0	0
figure_2	1	0	0	0	0	0
figure_3	1	0	0	0	0	0
table_a1	27	27	27	0	27	0
table_a2	27	27	27	0	27	0
table_a3	27	27	27	0	27	0
table_a4	27	27	27	0	27	0
table_b5	40	40	40	0	40	0
table_b6	40	40	40	0	40	0
table_b7	40	40	40	0	40	0
table_b8	40	40	40	0	40	0
table_c9	41	41	41	0	41	0
table_c10	41	41	41	0	41	0
text	25	0	0	0	21	0

The ground truth records 550 rows: every coefficient, standard error, sample size, log likelihood and AIC in Tables 2, 3 and 5 through 9 and in appendix Tables A.1 through C.10, the sixteen cell counts and the balance test of Table 3, the numbers the article states in prose on pages 13 through 17, and one row for each published float that carries no comparable number, so that a float’s absence can never be mistaken for a float that was checked and passed. Published values were extracted from the article and appendix PDFs by parsing the typeset tables rather than by transcription, so the comparison is exhaustive rather than selective; the extraction scripts are in `ground_truth/`.

38 published quantities are not printed by any deposited script: the 17 values of the Lucid column of Table 2, discussed below, and the 21 numbers the article states in prose, which the deposited scripts produce only as unlabelled cells inside larger tables.

`ground_truth/build_ground_truth.R` rebuilds the table at the end of every run. Its only hardcoded inputs are the published values, which serve as comparison targets; `value_rewrite` is read back out of `maintained/output/`, so the table cannot drift from the code it grades. Where a rewrite value disagreed with the article, `defect_locus` would record whether the fault lay with the article, the deposited code, the environment or this repository. No row needs it.

Five published floats carry no number that can be compared against the article, and each has a row saying why.

Table 4: Published floats with no comparable number.

Float	Why there is nothing in it to compare
table_1	Notation rather than results. The table sets out the potential outcomes of bilingual, Spanish-only, English-only and neither-language subjects under the control, English-ad and Spanish-ad conditions. Its only entries are two language indicators and six symbols; there is no estimated quantity in it.
table_4	The advertisement titles, running times, YouTube links and full transcripts in English and Spanish for all three candidates. No estimated quantity.

figure_1	The published figure prints no numbers, so there is nothing in it to compare against the article. The deposited script that draws it saves a PDF but prints and writes no estimate, so its estimates were recovered by refitting the models it specifies. All 20 plotted estimates agree with the rewrite on all four quantities, to within 0.
figure_2	As Figure 1. All 32 plotted estimates agree with the rewrite on all four quantities, to within 0.
figure_3	A calibration exercise rather than an estimate: the net effect of a Spanish-language strategy over the share of bilinguals in the electorate and the risk of mistargeting, holding the two effects at the values the article supposes on page 18. The surface is deterministic given those two numbers, and the rewrite writes 92 of its values to figure_3_simulation.csv. The deposited script draws the figure to the screen and never saves it, so the deposit contains no rendering of Figure 3 to compare against.

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity.

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_1	potential outcomes of four subject types					
table_2	lucid, age	34.8			34.8	1
table_2	lucid, age_se	0.3			0.3029	1
table_2	lucid, cuban	0.07			0.07411	1
table_2	lucid, cuban_se	0.01			0.006072	1
table_2	lucid, education_5	3.01			3.005	1
table_2	lucid, education_5_se	0.03			0.0251	1
table_2	lucid, female	0.7			0.7003	1
table_2	lucid, female_se	0.01			0.01062	1
table_2	lucid, income_7	4.11			4.109	1
table_2	lucid, income_7_se	0.05			0.05344	1
table_2	lucid, income_9	4.49			4.489	1
table_2	lucid, income_9_se	0.05			0.05204	1
table_2	lucid, mexican	0.49			0.4909	1
table_2	lucid, mexican_se	0.01			0.01159	1
table_2	lucid, n	1862			1862	1
table_2	lucid, other_hispanic	0.44			0.435	1
table_2	lucid, other_hispanic_se	0.01			0.01149	1
table_3	balance, chi_sq	7.3	7.258	1	7.258	1
table_3	balance, df	7	7	1	7	1
table_3	balance, p_value	0.4	0.4025	1	0.4025	1
table_3	experiment_1_bush_bilingual_english_survey, english_ad	462	462	1	462	1
table_3	experiment_1_bush_bilingual_english_survey, spanish_ad	488	488	1	488	1
table_3	experiment_1_bush_bilingual_spanish_survey, english_ad	452	452	1	452	1
table_3	experiment_1_bush_bilingual_spanish_survey, spanish_ad	460	460	1	460	1
table_3	experiment_2_vela_bilingual_english_survey, english_ad	437	437	1	437	1
table_3	experiment_2_vela_bilingual_english_survey, spanish_ad	442	442	1	442	1
table_3	experiment_2_vela_bilingual_spanish_survey, english_ad	420	420	1	420	1
table_3	experiment_2_vela_bilingual_spanish_survey, spanish_ad	382	382	1	382	1
table_3	experiment_2_vela_monolingual_english_survey, english_ad	675	675	1	675	1
table_3	experiment_2_vela_monolingual_english_survey, spanish_ad	669	669	1	669	1
table_3	experiment_3_coffman_bilingual_english_survey, english_ad	455	455	1	455	1
table_3	experiment_3_coffman_bilingual_english_survey, spanish_ad	424	424	1	424	1
table_3	experiment_3_coffman_bilingual_spanish_survey, english_ad	421	421	1	421	1
table_3	experiment_3_coffman_bilingual_spanish_survey, spanish_ad	381	381	1	381	1
table_3	experiment_3_coffman_monolingual_english_survey, english_ad	711	711	1	711	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_3	experiment_3_coffman_monolingual_english_survey, spanish_ad	633	633	1	633	1
table_4	advertisement treatments					
table_5	bush_bilingual, N	1849	1849	1	1849	1
table_5	bush_bilingual, Z_ad	0.049	0.049	1	0.04897	1
table_5	bush_bilingual, Z_ad_se	0.023	0.023	1	0.02322	1
table_5	bush_bilingual, Z_survey	0.005	0.005	1	0.005415	1
table_5	bush_bilingual, Z_survey_se	0.023	0.023	1	0.02323	1
table_5	bush_bilingual, constant	0.449	0.449	1	0.4495	1
table_5	bush_bilingual, constant_se	0.02	0.02	1	0.02009	1
table_5	coffman_bilingual, N	1681	1681	1	1681	1
table_5	coffman_bilingual, Z_ad	0.003	0.003	1	0.002706	1
table_5	coffman_bilingual, Z_ad_se	0.024	0.024	1	0.02362	1
table_5	coffman_bilingual, Z_survey	-0.028	-0.028	1	-0.02779	1
table_5	coffman_bilingual, Z_survey_se	0.024	0.024	1	0.02361	1
table_5	coffman_bilingual, constant	0.384	0.384	1	0.3844	1
table_5	coffman_bilingual, constant_se	0.02	0.02	1	0.02013	1
table_5	coffman_monolingual, N	1343	1343	1	1343	1
table_5	coffman_monolingual, Z_ad	-0.187	-0.187	1	-0.1872	1
table_5	coffman_monolingual, Z_ad_se	0.026	0.026	1	0.02645	1
table_5	coffman_monolingual, constant	0.513	0.513	1	0.5127	1
table_5	coffman_monolingual, constant_se	0.019	0.019	1	0.01877	1
table_5	vela_bilingual, N	1680	1680	1	1680	1
table_5	vela_bilingual, Z_ad	0.049	0.049	1	0.04855	1
table_5	vela_bilingual, Z_ad_se	0.024	0.024	1	0.0239	1
table_5	vela_bilingual, Z_survey	0.073	0.073	1	0.07296	1
table_5	vela_bilingual, Z_survey_se	0.024	0.024	1	0.0239	1
table_5	vela_bilingual, constant	0.535	0.535	1	0.5348	1
table_5	vela_bilingual, constant_se	0.021	0.021	1	0.02072	1
table_5	vela_monolingual, N	1343	1343	1	1343	1
table_5	vela_monolingual, Z_ad	-0.02	-0.02	1	-0.02012	1
table_5	vela_monolingual, Z_ad_se	0.026	0.026	1	0.02614	1
table_5	vela_monolingual, constant	0.366	0.366	1	0.3659	1
table_5	vela_monolingual, constant_se	0.019	0.019	1	0.01855	1
table_6	bush_bilingual, N	1862	1862	1	1862	1
table_6	bush_bilingual, Z_ad	0.167	0.167	1	0.1668	1
table_6	bush_bilingual, Z_ad_se	0.075	0.075	1	0.07533	1
table_6	bush_bilingual, Z_survey	0.165	0.165	1	0.165	1
table_6	bush_bilingual, Z_survey_se	0.075	0.075	1	0.07546	1
table_6	bush_bilingual, constant	4.81	4.81	1	4.81	1
table_6	bush_bilingual, constant_se	0.063	0.063	1	0.06343	1
table_6	coffman_bilingual, N	1680	1680	1	1680	1
table_6	coffman_bilingual, Z_ad	0.019	0.019	1	0.01945	1
table_6	coffman_bilingual, Z_ad_se	0.077	0.077	1	0.07728	1
table_6	coffman_bilingual, Z_survey	0.093	0.093	1	0.09271	1
table_6	coffman_bilingual, Z_survey_se	0.078	0.078	1	0.07773	1
table_6	coffman_bilingual, constant	4.902	4.902	1	4.902	1
table_6	coffman_bilingual, constant_se	0.066	0.066	1	0.06603	1
table_6	coffman_monolingual, N	1342	1342	1	1342	1
table_6	coffman_monolingual, Z_ad	-0.83	-0.83	1	-0.8299	1
table_6	coffman_monolingual, Z_ad_se	0.074	0.074	1	0.07376	1
table_6	coffman_monolingual, constant	5.161	5.161	1	5.161	1
table_6	coffman_monolingual, constant_se	0.054	0.054	1	0.05444	1
table_6	vela_bilingual, N	1680	1680	1	1680	1
table_6	vela_bilingual, Z_ad	0.068	0.068	1	0.0676	1
table_6	vela_bilingual, Z_ad_se	0.067	0.067	1	0.06747	1
table_6	vela_bilingual, Z_survey	0.149	0.149	1	0.1494	1
table_6	vela_bilingual, Z_survey_se	0.067	0.067	1	0.06745	1
table_6	vela_bilingual, constant	4.994	4.994	1	4.994	1
table_6	vela_bilingual, constant_se	0.057	0.057	1	0.05685	1
table_6	vela_monolingual, N	1341	1341	1	1341	1
table_6	vela_monolingual, Z_ad	-0.472	-0.472	1	-0.4723	1
table_6	vela_monolingual, Z_ad_se	0.07	0.07	1	0.06956	1
table_6	vela_monolingual, constant	4.712	4.712	1	4.712	1
table_6	vela_monolingual, constant_se	0.051	0.051	1	0.05075	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_7	bush_bilingual, N	1858	1858	1	1858	1
table_7	bush_bilingual, Z_ad	0.037	0.037	1	0.03737	1
table_7	bush_bilingual, Z_ad_se	0.02	0.02	1	0.02045	1
table_7	bush_bilingual, Z_survey	-0.035	-0.035	1	-0.03475	1
table_7	bush_bilingual, Z_survey_se	0.02	0.02	1	0.02045	1
table_7	bush_bilingual, constant	0.734	0.734	1	0.7342	1
table_7	bush_bilingual, constant_se	0.018	0.018	1	0.01776	1
table_7	coffman_bilingual, N	1680	1680	1	1680	1
table_7	coffman_bilingual, Z_ad	0.035	0.035	1	0.03456	1
table_7	coffman_bilingual, Z_ad_se	0.021	0.021	1	0.02138	1
table_7	coffman_bilingual, Z_survey	-0.045	-0.045	1	-0.04488	1
table_7	coffman_bilingual, Z_survey_se	0.021	0.021	1	0.02148	1
table_7	coffman_bilingual, constant	0.744	0.744	1	0.7442	1
table_7	coffman_bilingual, constant_se	0.018	0.018	1	0.01812	1
table_7	coffman_monolingual, N	1337	1337	1	1337	1
table_7	coffman_monolingual, Z_ad	-0.155	-0.155	1	-0.1552	1
table_7	coffman_monolingual, Z_ad_se	0.024	0.024	1	0.02389	1
table_7	coffman_monolingual, constant	0.815	0.815	1	0.815	1
table_7	coffman_monolingual, constant_se	0.015	0.015	1	0.0146	1
table_7	vela_bilingual, N	1680	1680	1	1680	1
table_7	vela_bilingual, Z_ad	0.018	0.018	1	0.01761	1
table_7	vela_bilingual, Z_ad_se	0.017	0.017	1	0.01682	1
table_7	vela_bilingual, Z_survey	-0.024	-0.024	1	-0.02399	1
table_7	vela_bilingual, Z_survey_se	0.017	0.017	1	0.01691	1
table_7	vela_bilingual, constant	0.865	0.865	1	0.8647	1
table_7	vela_bilingual, constant_se	0.014	0.014	1	0.01425	1
table_7	vela_monolingual, N	1336	1336	1	1336	1
table_7	vela_monolingual, Z_ad	-0.152	-0.152	1	-0.152	1
table_7	vela_monolingual, Z_ad_se	0.025	0.025	1	0.02466	1
table_7	vela_monolingual, constant	0.78	0.78	1	0.7804	1
table_7	vela_monolingual, constant_se	0.016	0.016	1	0.01596	1
table_8	bush_bilingual, N	1861	1861	1	1861	1
table_8	bush_bilingual, Z_ad	0.076	0.076	1	0.07579	1
table_8	bush_bilingual, Z_ad_se	0.042	0.042	1	0.04152	1
table_8	bush_bilingual, Z_survey	-0.018	-0.018	1	-0.01848	1
table_8	bush_bilingual, Z_survey_se	0.042	0.042	1	0.04156	1
table_8	bush_bilingual, constant	1.768	1.768	1	1.768	1
table_8	bush_bilingual, constant_se	0.035	0.035	1	0.03545	1
table_8	coffman_bilingual, N	1679	1679	1	1679	1
table_8	coffman_bilingual, Z_ad	0.03	0.03	1	0.0304	1
table_8	coffman_bilingual, Z_ad_se	0.041	0.041	1	0.0409	1
table_8	coffman_bilingual, Z_survey	-0.084	-0.084	1	-0.08435	1
table_8	coffman_bilingual, Z_survey_se	0.041	0.041	1	0.04092	1
table_8	coffman_bilingual, constant	1.814	1.814	1	1.814	1
table_8	coffman_bilingual, constant_se	0.034	0.034	1	0.03399	1
table_8	coffman_monolingual, N	1338	1338	1	1338	1
table_8	coffman_monolingual, Z_ad	-0.31	-0.31	1	-0.3101	1
table_8	coffman_monolingual, Z_ad_se	0.044	0.044	1	0.0441	1
table_8	coffman_monolingual, constant	1.966	1.966	1	1.966	1
table_8	coffman_monolingual, constant_se	0.029	0.029	1	0.02947	1
table_8	vela_bilingual, N	1680	1680	1	1680	1
table_8	vela_bilingual, Z_ad	0.103	0.103	1	0.1027	1
table_8	vela_bilingual, Z_ad_se	0.038	0.038	1	0.03787	1
table_8	vela_bilingual, Z_survey	0.011	0.011	1	0.01076	1
table_8	vela_bilingual, Z_survey_se	0.038	0.038	1	0.03793	1
table_8	vela_bilingual, constant	1.915	1.915	1	1.915	1
table_8	vela_bilingual, constant_se	0.033	0.033	1	0.03254	1
table_8	vela_monolingual, N	1338	1338	1	1338	1
table_8	vela_monolingual, Z_ad	-0.16	-0.16	1	-0.1602	1
table_8	vela_monolingual, Z_ad_se	0.047	0.047	1	0.04695	1
table_8	vela_monolingual, constant	1.73	1.73	1	1.73	1
table_8	vela_monolingual, constant_se	0.031	0.031	1	0.03141	1
table_9	bush_bilingual, N	1861	1861	1	1861	1
table_9	bush_bilingual, Z_survey	0.243	0.243	1	0.2427	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_9	bush_bilingual, Z_survey_se	0.04	0.04	1	0.0403	1
table_9	bush_bilingual, constant	2.08	2.08	1	2.08	1
table_9	bush_bilingual, constant_se	0.031	0.031	1	0.03121	1
table_9	vela_coffman_bilingual, N	1681	1681	1	1681	1
table_9	vela_coffman_bilingual, Z_survey	0.131	0.131	1	0.1311	1
table_9	vela_coffman_bilingual, Z_survey_se	0.041	0.041	1	0.04127	1
table_9	vela_coffman_bilingual, constant	2.106	2.106	1	2.106	1
table_9	vela_coffman_bilingual, constant_se	0.031	0.031	1	0.03099	1
figure_1	all plotted estimates, estimate, standard error and both confidence limits					
figure_2	all plotted estimates, estimate, standard error and both confidence limits					
figure_3	net effect surface					
table_a1	bush_bilingual, N	1849	1849	1	1849	1
table_a1	bush_bilingual, Z_ad	0.061	0.061	1	0.06065	1
table_a1	bush_bilingual, Z_ad_se	0.032	0.032	1	0.03248	1
table_a1	bush_bilingual, Z_survey	0.018	0.018	1	0.01755	1
table_a1	bush_bilingual, Z_survey_se	0.033	0.033	1	0.03305	1
table_a1	bush_bilingual, constant	0.443	0.443	1	0.4435	1
table_a1	bush_bilingual, constant_se	0.023	0.023	1	0.02319	1
table_a1	bush_bilingual, interaction	-0.024	-0.024	1	-0.02387	1
table_a1	bush_bilingual, interaction_se	0.046	0.046	1	0.04647	1
table_a1	coffman_bilingual, N	1681	1681	1	1681	1
table_a1	coffman_bilingual, Z_ad	-0.039	-0.039	1	-0.03883	1
table_a1	coffman_bilingual, Z_ad_se	0.033	0.033	1	0.03285	1
table_a1	coffman_bilingual, Z_survey	-0.069	-0.069	1	-0.06948	1
table_a1	coffman_bilingual, Z_survey_se	0.033	0.033	1	0.03257	1
table_a1	coffman_bilingual, constant	0.404	0.404	1	0.4044	1
table_a1	coffman_bilingual, constant_se	0.023	0.023	1	0.02303	1
table_a1	coffman_bilingual, interaction	0.087	0.087	1	0.08711	1
table_a1	coffman_bilingual, interaction_se	0.047	0.047	1	0.04724	1
table_a1	vela_bilingual, N	1680	1680	1	1680	1
table_a1	vela_bilingual, Z_ad	0.075	0.075	1	0.07463	1
table_a1	vela_bilingual, Z_ad_se	0.033	0.033	1	0.03346	1
table_a1	vela_bilingual, Z_survey	0.1	0.1	1	0.09969	1
table_a1	vela_bilingual, Z_survey_se	0.034	0.034	1	0.03367	1
table_a1	vela_bilingual, constant	0.522	0.522	1	0.5217	1
table_a1	vela_bilingual, constant_se	0.024	0.024	1	0.02392	1
table_a1	vela_bilingual, interaction	-0.055	-0.055	1	-0.0547	1
table_a1	vela_bilingual, interaction_se	0.048	0.048	1	0.0478	1
table_a2	bush_bilingual, N	1858	1858	1	1858	1
table_a2	bush_bilingual, Z_ad	0.034	0.034	1	0.03409	1
table_a2	bush_bilingual, Z_ad_se	0.028	0.028	1	0.02803	1
table_a2	bush_bilingual, Z_survey	-0.038	-0.038	1	-0.03815	1
table_a2	bush_bilingual, Z_survey_se	0.03	0.03	1	0.02985	1
table_a2	bush_bilingual, constant	0.736	0.736	1	0.7359	1
table_a2	bush_bilingual, constant_se	0.021	0.021	1	0.02053	1
table_a2	bush_bilingual, interaction	0.007	0.007	1	0.006695	1
table_a2	bush_bilingual, interaction_se	0.041	0.041	1	0.04095	1
table_a2	coffman_bilingual, N	1680	1680	1	1680	1
table_a2	coffman_bilingual, Z_ad	0.051	0.051	1	0.05097	1
table_a2	coffman_bilingual, Z_ad_se	0.029	0.029	1	0.02872	1
table_a2	coffman_bilingual, Z_survey	-0.028	-0.028	1	-0.02843	1
table_a2	coffman_bilingual, Z_survey_se	0.03	0.03	1	0.03033	1
table_a2	coffman_bilingual, constant	0.736	0.736	1	0.7363	1
table_a2	coffman_bilingual, constant_se	0.021	0.021	1	0.02068	1
table_a2	coffman_bilingual, interaction	-0.034	-0.034	1	-0.0344	1
table_a2	coffman_bilingual, interaction_se	0.043	0.043	1	0.04292	1
table_a2	vela_bilingual, N	1680	1680	1	1680	1
table_a2	vela_bilingual, Z_ad	0.013	0.013	1	0.01254	1
table_a2	vela_bilingual, Z_ad_se	0.022	0.022	1	0.02246	1
table_a2	vela_bilingual, Z_survey	-0.029	-0.029	1	-0.02918	1
table_a2	vela_bilingual, Z_survey_se	0.024	0.024	1	0.02425	1
table_a2	vela_bilingual, constant	0.867	0.867	1	0.8673	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_a2	vela_bilingual, constant_se	0.016	0.016	1	0.01625	1
table_a2	vela_bilingual, interaction	0.011	0.011	1	0.01062	1
table_a2	vela_bilingual, interaction_se	0.034	0.034	1	0.03379	1
table_a3	bush_bilingual, N	1861	1861	1	1861	1
table_a3	bush_bilingual, Z_ad	0.135	0.135	1	0.1349	1
table_a3	bush_bilingual, Z_ad_se	0.058	0.058	1	0.05766	1
table_a3	bush_bilingual, Z_survey	0.043	0.043	1	0.04288	1
table_a3	bush_bilingual, Z_survey_se	0.058	0.058	1	0.05816	1
table_a3	bush_bilingual, constant	1.738	1.738	1	1.738	1
table_a3	bush_bilingual, constant_se	0.041	0.041	1	0.04061	1
table_a3	bush_bilingual, interaction	-0.121	-0.121	1	-0.1206	1
table_a3	bush_bilingual, interaction_se	0.083	0.083	1	0.08305	1
table_a3	coffman_bilingual, N	1679	1679	1	1679	1
table_a3	coffman_bilingual, Z_ad	-0.04	-0.04	1	-0.03951	1
table_a3	coffman_bilingual, Z_ad_se	0.056	0.056	1	0.0562	1
table_a3	coffman_bilingual, Z_survey	-0.154	-0.154	1	-0.1544	1
table_a3	coffman_bilingual, Z_survey_se	0.057	0.057	1	0.05654	1
table_a3	coffman_bilingual, constant	1.848	1.848	1	1.848	1
table_a3	coffman_bilingual, constant_se	0.038	0.038	1	0.03848	1
table_a3	coffman_bilingual, interaction	0.146	0.146	1	0.1464	1
table_a3	coffman_bilingual, interaction_se	0.082	0.082	1	0.08188	1
table_a3	vela_bilingual, N	1680	1680	1	1680	1
table_a3	vela_bilingual, Z_ad	0.107	0.107	1	0.1074	1
table_a3	vela_bilingual, Z_ad_se	0.052	0.052	1	0.05209	1
table_a3	vela_bilingual, Z_survey	0.016	0.016	1	0.01553	1
table_a3	vela_bilingual, Z_survey_se	0.054	0.054	1	0.05353	1
table_a3	vela_bilingual, constant	1.913	1.913	1	1.913	1
table_a3	vela_bilingual, constant_se	0.037	0.037	1	0.03741	1
table_a3	vela_bilingual, interaction	-0.01	-0.01	1	-0.009758	1
table_a3	vela_bilingual, interaction_se	0.076	0.076	1	0.07587	1
table_a4	bush_bilingual, N	1862	1862	1	1862	1
table_a4	bush_bilingual, Z_ad	0.248	0.248	1	0.248	1
table_a4	bush_bilingual, Z_ad_se	0.101	0.101	1	0.1014	1
table_a4	bush_bilingual, Z_survey	0.249	0.249	1	0.2493	1
table_a4	bush_bilingual, Z_survey_se	0.107	0.107	1	0.1072	1
table_a4	bush_bilingual, constant	4.768	4.768	1	4.768	1
table_a4	bush_bilingual, constant_se	0.072	0.072	1	0.07207	1
table_a4	bush_bilingual, interaction	-0.166	-0.166	1	-0.1657	1
table_a4	bush_bilingual, interaction_se	0.151	0.151	1	0.1509	1
table_a4	coffman_bilingual, N	1680	1680	1	1680	1
table_a4	coffman_bilingual, Z_ad	-0.089	-0.089	1	-0.0886	1
table_a4	coffman_bilingual, Z_ad_se	0.104	0.104	1	0.1036	1
table_a4	coffman_bilingual, Z_survey	-0.016	-0.016	1	-0.0156	1
table_a4	coffman_bilingual, Z_survey_se	0.111	0.111	1	0.1108	1
table_a4	coffman_bilingual, constant	4.954	4.954	1	4.954	1
table_a4	coffman_bilingual, constant_se	0.076	0.076	1	0.07558	1
table_a4	coffman_bilingual, interaction	0.226	0.226	1	0.2265	1
table_a4	coffman_bilingual, interaction_se	0.155	0.155	1	0.1551	1
table_a4	vela_bilingual, N	1680	1680	1	1680	1
table_a4	vela_bilingual, Z_ad	0.166	0.166	1	0.1658	1
table_a4	vela_bilingual, Z_ad_se	0.093	0.093	1	0.09341	1
table_a4	vela_bilingual, Z_survey	0.25	0.25	1	0.2502	1
table_a4	vela_bilingual, Z_survey_se	0.092	0.092	1	0.0925	1
table_a4	vela_bilingual, constant	4.945	4.945	1	4.945	1
table_a4	vela_bilingual, constant_se	0.065	0.065	1	0.06474	1
table_a4	vela_bilingual, interaction	-0.206	-0.206	1	-0.2062	1
table_a4	vela_bilingual, interaction_se	0.135	0.135	1	0.135	1
table_b5	coffman_bilingual_democrat, N	1029	1029	1	1029	1
table_b5	coffman_bilingual_democrat, Z_ad	-0.003	-0.003	1	-0.002698	1
table_b5	coffman_bilingual_democrat, Z_ad_se	0.027	0.027	1	0.02693	1
table_b5	coffman_bilingual_democrat, constant	0.248	0.248	1	0.2481	1
table_b5	coffman_bilingual_democrat, constant_se	0.019	0.019	1	0.01867	1
table_b5	coffman_bilingual_republican, N	335	335	1	335	1
table_b5	coffman_bilingual_republican, Z_ad	0.022	0.022	1	0.02232	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_b5	coffman_bilingual_republican, Z_ad_se	0.045	0.045	1	0.0452	1
table_b5	coffman_bilingual_republican, constant	0.771	0.771	1	0.7714	1
table_b5	coffman_bilingual_republican, constant_se	0.032	0.032	1	0.03183	1
table_b5	coffman_monolingual_democrat, N	579	579	1	579	1
table_b5	coffman_monolingual_democrat, Z_ad	-0.141	-0.141	1	-0.1405	1
table_b5	coffman_monolingual_democrat, Z_ad_se	0.033	0.033	1	0.0334	1
table_b5	coffman_monolingual_democrat, constant	0.281	0.281	1	0.2814	1
table_b5	coffman_monolingual_democrat, constant_se	0.026	0.026	1	0.02622	1
table_b5	coffman_monolingual_republican, N	491	491	1	491	1
table_b5	coffman_monolingual_republican, Z_ad	-0.168	-0.168	1	-0.1678	1
table_b5	coffman_monolingual_republican, Z_ad_se	0.04	0.04	1	0.03999	1
table_b5	coffman_monolingual_republican, constant	0.816	0.816	1	0.8162	1
table_b5	coffman_monolingual_republican, constant_se	0.024	0.024	1	0.02353	1
table_b5	vela_bilingual_democrat, N	1029	1029	1	1029	1
table_b5	vela_bilingual_democrat, Z_ad	0.064	0.064	1	0.06365	1
table_b5	vela_bilingual_democrat, Z_ad_se	0.026	0.026	1	0.02592	1
table_b5	vela_bilingual_democrat, constant	0.744	0.744	1	0.7443	1
table_b5	vela_bilingual_democrat, constant_se	0.019	0.019	1	0.01908	1
table_b5	vela_bilingual_republican, N	335	335	1	335	1
table_b5	vela_bilingual_republican, Z_ad	-0.01	-0.01	1	-0.01023	1
table_b5	vela_bilingual_republican, Z_ad_se	0.05	0.05	1	0.04955	1
table_b5	vela_bilingual_republican, constant	0.292	0.292	1	0.2917	1
table_b5	vela_bilingual_republican, constant_se	0.035	0.035	1	0.03517	1
table_b5	vela_monolingual_democrat, N	579	579	1	579	1
table_b5	vela_monolingual_democrat, Z_ad	-0.044	-0.044	1	-0.04366	1
table_b5	vela_monolingual_democrat, Z_ad_se	0.04	0.04	1	0.04025	1
table_b5	vela_monolingual_democrat, constant	0.648	0.648	1	0.6477	1
table_b5	vela_monolingual_democrat, constant_se	0.029	0.029	1	0.02855	1
table_b5	vela_monolingual_republican, N	491	491	1	491	1
table_b5	vela_monolingual_republican, Z_ad	-0.011	-0.011	1	-0.01121	1
table_b5	vela_monolingual_republican, Z_ad_se	0.031	0.031	1	0.03063	1
table_b5	vela_monolingual_republican, constant	0.138	0.138	1	0.1378	1
table_b5	vela_monolingual_republican, constant_se	0.022	0.022	1	0.02167	1
table_b6	coffman_bilingual_democrat, N	1029	1029	1	1029	1
table_b6	coffman_bilingual_democrat, Z_ad	0.146	0.146	1	0.1461	1
table_b6	coffman_bilingual_democrat, Z_ad_se	0.102	0.102	1	0.1024	1
table_b6	coffman_bilingual_democrat, constant	4.722	4.722	1	4.722	1
table_b6	coffman_bilingual_democrat, constant_se	0.072	0.072	1	0.07227	1
table_b6	coffman_bilingual_republican, N	335	335	1	335	1
table_b6	coffman_bilingual_republican, Z_ad	-0.303	-0.303	1	-0.303	1
table_b6	coffman_bilingual_republican, Z_ad_se	0.159	0.159	1	0.1593	1
table_b6	coffman_bilingual_republican, constant	5.634	5.634	1	5.634	1
table_b6	coffman_bilingual_republican, constant_se	0.112	0.112	1	0.1115	1
table_b6	coffman_monolingual_democrat, N	579	579	1	579	1
table_b6	coffman_monolingual_democrat, Z_ad	-0.702	-0.702	1	-0.7019	1
table_b6	coffman_monolingual_democrat, Z_ad_se	0.114	0.114	1	0.1141	1
table_b6	coffman_monolingual_democrat, constant	4.969	4.969	1	4.969	1
table_b6	coffman_monolingual_democrat, constant_se	0.086	0.086	1	0.08582	1
table_b6	coffman_monolingual_republican, N	490	490	1	490	1
table_b6	coffman_monolingual_republican, Z_ad	-1.004	-1.004	1	-1.004	1
table_b6	coffman_monolingual_republican, Z_ad_se	0.123	0.123	1	0.1227	1
table_b6	coffman_monolingual_republican, constant	5.467	5.467	1	5.467	1
table_b6	coffman_monolingual_republican, constant_se	0.083	0.083	1	0.08332	1
table_b6	vela_bilingual_democrat, N	1028	1028	1	1028	1
table_b6	vela_bilingual_democrat, Z_ad	0.116	0.116	1	0.1157	1
table_b6	vela_bilingual_democrat, Z_ad_se	0.082	0.082	1	0.08197	1
table_b6	vela_bilingual_democrat, constant	5.317	5.317	1	5.317	1
table_b6	vela_bilingual_democrat, constant_se	0.057	0.057	1	0.05735	1
table_b6	vela_bilingual_republican, N	335	335	1	335	1
table_b6	vela_bilingual_republican, Z_ad	-0.247	-0.247	1	-0.2468	1
table_b6	vela_bilingual_republican, Z_ad_se	0.16	0.16	1	0.1599	1
table_b6	vela_bilingual_republican, constant	4.792	4.792	1	4.792	1
table_b6	vela_bilingual_republican, constant_se	0.109	0.109	1	0.1094	1
table_b6	vela_monolingual_democrat, N	578	578	1	578	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_b6	vela_monolingual_democrat, Z_ad	-0.608	-0.608	1	-0.6084	1
table_b6	vela_monolingual_democrat, Z_ad_se	0.102	0.102	1	0.1024	1
table_b6	vela_monolingual_democrat, constant	5.129	5.129	1	5.129	1
table_b6	vela_monolingual_democrat, constant_se	0.078	0.078	1	0.07768	1
table_b6	vela_monolingual_republican, N	491	491	1	491	1
table_b6	vela_monolingual_republican, Z_ad	-0.399	-0.399	1	-0.399	1
table_b6	vela_monolingual_republican, Z_ad_se	0.118	0.118	1	0.1184	1
table_b6	vela_monolingual_republican, constant	4.437	4.437	1	4.437	1
table_b6	vela_monolingual_republican, constant_se	0.081	0.081	1	0.08147	1
table_b7	coffman_bilingual_democrat, N	1029	1029	1	1029	1
table_b7	coffman_bilingual_democrat, Z_ad	0.078	0.078	1	0.07805	1
table_b7	coffman_bilingual_democrat, Z_ad_se	0.028	0.028	1	0.02844	1
table_b7	coffman_bilingual_democrat, constant	0.662	0.662	1	0.6623	1
table_b7	coffman_bilingual_democrat, constant_se	0.02	0.02	1	0.02045	1
table_b7	coffman_bilingual_republican, N	335	335	1	335	1
table_b7	coffman_bilingual_republican, Z_ad	-0.055	-0.055	1	-0.055	1
table_b7	coffman_bilingual_republican, Z_ad_se	0.039	0.039	1	0.03892	1
table_b7	coffman_bilingual_republican, constant	0.88	0.88	1	0.88	1
table_b7	coffman_bilingual_republican, constant_se	0.025	0.025	1	0.02464	1
table_b7	coffman_monolingual_democrat, N	575	575	1	575	1
table_b7	coffman_monolingual_democrat, Z_ad	-0.133	-0.133	1	-0.1326	1
table_b7	coffman_monolingual_democrat, Z_ad_se	0.037	0.037	1	0.03735	1
table_b7	coffman_monolingual_democrat, constant	0.782	0.782	1	0.7816	1
table_b7	coffman_monolingual_democrat, constant_se	0.024	0.024	1	0.02418	1
table_b7	coffman_monolingual_republican, N	490	490	1	490	1
table_b7	coffman_monolingual_republican, Z_ad	-0.181	-0.181	1	-0.1814	1
table_b7	coffman_monolingual_republican, Z_ad_se	0.037	0.037	1	0.03696	1
table_b7	coffman_monolingual_republican, constant	0.879	0.879	1	0.8787	1
table_b7	coffman_monolingual_republican, constant_se	0.02	0.02	1	0.01983	1
table_b7	vela_bilingual_democrat, N	1029	1029	1	1029	1
table_b7	vela_bilingual_democrat, Z_ad	0.04	0.04	1	0.03962	1
table_b7	vela_bilingual_democrat, Z_ad_se	0.017	0.017	1	0.01734	1
table_b7	vela_bilingual_democrat, constant	0.895	0.895	1	0.895	1
table_b7	vela_bilingual_democrat, constant_se	0.013	0.013	1	0.0134	1
table_b7	vela_bilingual_republican, N	335	335	1	335	1
table_b7	vela_bilingual_republican, Z_ad	-0.049	-0.049	1	-0.04926	1
table_b7	vela_bilingual_republican, Z_ad_se	0.047	0.047	1	0.04744	1
table_b7	vela_bilingual_republican, constant	0.774	0.774	1	0.7738	1
table_b7	vela_bilingual_republican, constant_se	0.032	0.032	1	0.03237	1
table_b7	vela_monolingual_democrat, N	576	576	1	576	1
table_b7	vela_monolingual_democrat, Z_ad	-0.113	-0.113	1	-0.1126	1
table_b7	vela_monolingual_democrat, Z_ad_se	0.032	0.032	1	0.03195	1
table_b7	vela_monolingual_democrat, constant	0.872	0.872	1	0.8719	1
table_b7	vela_monolingual_democrat, constant_se	0.02	0.02	1	0.01997	1
table_b7	vela_monolingual_republican, N	490	490	1	490	1
table_b7	vela_monolingual_republican, Z_ad	-0.226	-0.226	1	-0.2256	1
table_b7	vela_monolingual_republican, Z_ad_se	0.043	0.043	1	0.04334	1
table_b7	vela_monolingual_republican, constant	0.709	0.709	1	0.7087	1
table_b7	vela_monolingual_republican, constant_se	0.029	0.029	1	0.02857	1
table_b8	coffman_bilingual_democrat, N	1028	1028	1	1028	1
table_b8	coffman_bilingual_democrat, Z_ad	0.128	0.128	1	0.1284	1
table_b8	coffman_bilingual_democrat, Z_ad_se	0.052	0.052	1	0.05224	1
table_b8	coffman_bilingual_democrat, constant	1.65	1.65	1	1.65	1
table_b8	coffman_bilingual_democrat, constant_se	0.036	0.036	1	0.03626	1
table_b8	coffman_bilingual_republican, N	335	335	1	335	1
table_b8	coffman_bilingual_republican, Z_ad	-0.139	-0.139	1	-0.1391	1
table_b8	coffman_bilingual_republican, Z_ad_se	0.084	0.084	1	0.08357	1
table_b8	coffman_bilingual_republican, constant	2.183	2.183	1	2.183	1
table_b8	coffman_bilingual_republican, constant_se	0.051	0.051	1	0.0513	1
table_b8	coffman_monolingual_democrat, N	577	577	1	577	1
table_b8	coffman_monolingual_democrat, Z_ad	-0.179	-0.179	1	-0.1786	1
table_b8	coffman_monolingual_democrat, Z_ad_se	0.067	0.067	1	0.06671	1
table_b8	coffman_monolingual_democrat, constant	1.895	1.895	1	1.895	1
table_b8	coffman_monolingual_democrat, constant_se	0.047	0.047	1	0.04677	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_b8	coffman_monolingual_republican, N	489	489	1	489	1
table_b8	coffman_monolingual_republican, Z_ad	-0.491	-0.491	1	-0.491	1
table_b8	coffman_monolingual_republican, Z_ad_se	0.073	0.073	1	0.07269	1
table_b8	coffman_monolingual_republican, constant	2.099	2.099	1	2.099	1
table_b8	coffman_monolingual_republican, constant_se	0.043	0.043	1	0.04332	1
table_b8	vela_bilingual_democrat, N	1029	1029	1	1029	1
table_b8	vela_bilingual_democrat, Z_ad	0.11	0.11	1	0.1098	1
table_b8	vela_bilingual_democrat, Z_ad_se	0.042	0.042	1	0.04218	1
table_b8	vela_bilingual_democrat, constant	2.076	2.076	1	2.076	1
table_b8	vela_bilingual_democrat, constant_se	0.029	0.029	1	0.02915	1
table_b8	vela_bilingual_republican, N	335	335	1	335	1
table_b8	vela_bilingual_republican, Z_ad	-0.073	-0.073	1	-0.07339	1
table_b8	vela_bilingual_republican, Z_ad_se	0.096	0.096	1	0.09595	1
table_b8	vela_bilingual_republican, constant	1.744	1.744	1	1.744	1
table_b8	vela_bilingual_republican, constant_se	0.071	0.071	1	0.07064	1
table_b8	vela_monolingual_democrat, N	577	577	1	577	1
table_b8	vela_monolingual_democrat, Z_ad	-0.181	-0.181	1	-0.1807	1
table_b8	vela_monolingual_democrat, Z_ad_se	0.062	0.062	1	0.06249	1
table_b8	vela_monolingual_democrat, constant	2.032	2.032	1	2.032	1
table_b8	vela_monolingual_democrat, constant_se	0.041	0.041	1	0.04138	1
table_b8	vela_monolingual_republican, N	490	490	1	490	1
table_b8	vela_monolingual_republican, Z_ad	-0.192	-0.192	1	-0.1922	1
table_b8	vela_monolingual_republican, Z_ad_se	0.079	0.079	1	0.07948	1
table_b8	vela_monolingual_republican, constant	1.48	1.48	1	1.48	1
table_b8	vela_monolingual_republican, constant_se	0.052	0.052	1	0.05165	1
table_c9	bush_bilingual, N	1849	1849	1	1849	1
table_c9	bush_bilingual, Z_ad	0.196	0.196	1	0.1965	1
table_c9	bush_bilingual, Z_ad_se	0.093	0.093	1	0.09326	1
table_c9	bush_bilingual, Z_survey	0.022	0.022	1	0.02176	1
table_c9	bush_bilingual, Z_survey_se	0.093	0.093	1	0.09326	1
table_c9	bush_bilingual, aic	2561	2561	1	2561	1
table_c9	bush_bilingual, constant	-0.203	-0.203	1	-0.2028	1
table_c9	bush_bilingual, constant_se	0.081	0.081	1	0.08104	1
table_c9	bush_bilingual, log_likelihood	-1277	-1277	1	-1277	1
table_c9	coffman_bilingual, N	1681	1681	1	1681	1
table_c9	coffman_bilingual, Z_ad	0.012	0.012	1	0.01158	1
table_c9	coffman_bilingual, Z_ad_se	0.101	0.101	1	0.101	1
table_c9	coffman_bilingual, Z_survey	-0.119	-0.119	1	-0.119	1
table_c9	coffman_bilingual, Z_survey_se	0.101	0.101	1	0.1011	1
table_c9	coffman_bilingual, aic	2224	2224	1	2224	1
table_c9	coffman_bilingual, constant	-0.471	-0.471	1	-0.4712	1
table_c9	coffman_bilingual, constant_se	0.085	0.085	1	0.08475	1
table_c9	coffman_bilingual, log_likelihood	-1109	-1109	1	-1109	1
table_c9	coffman_monolingual, N	1343	1343	1	1343	1
table_c9	coffman_monolingual, Z_ad	-0.78	-0.78	1	-0.7796	1
table_c9	coffman_monolingual, Z_ad_se	0.113	0.113	1	0.1133	1
table_c9	coffman_monolingual, aic	1787	1787	1	1787	1
table_c9	coffman_monolingual, constant	0.051	0.051	1	0.05072	1
table_c9	coffman_monolingual, constant_se	0.075	0.075	1	0.07508	1
table_c9	coffman_monolingual, log_likelihood	-891.3	-891.3	1	-891.3	1
table_c9	vela_bilingual, N	1680	1680	1	1680	1
table_c9	vela_bilingual, Z_ad	0.203	0.203	1	0.2026	1
table_c9	vela_bilingual, Z_ad_se	0.1	0.1	1	0.09986	1
table_c9	vela_bilingual, Z_survey	0.304	0.304	1	0.3042	1
table_c9	vela_bilingual, Z_survey_se	0.1	0.1	1	0.1001	1
table_c9	vela_bilingual, aic	2263	2263	1	2263	1
table_c9	vela_bilingual, constant	0.137	0.137	1	0.1369	1
table_c9	vela_bilingual, constant_se	0.084	0.084	1	0.08419	1
table_c9	vela_bilingual, log_likelihood	-1128	-1128	1	-1128	1
table_c9	vela_monolingual, N	1343	1343	1	1343	1
table_c9	vela_monolingual, Z_ad	-0.088	-0.088	1	-0.08778	1
table_c9	vela_monolingual, Z_ad_se	0.114	0.114	1	0.114	1
table_c9	vela_monolingual, aic	1752	1752	1	1752	1
table_c9	vela_monolingual, constant	-0.55	-0.55	1	-0.5497	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_c9	vela_monolingual, constant_se	0.08	0.08	1	0.07991	1
table_c9	vela_monolingual, log_likelihood	-874	-874	1	-874	1
table_c10	bush_bilingual, N	1858	1858	1	1858	1
table_c10	bush_bilingual, Z_ad	0.193	0.193	1	0.1928	1
table_c10	bush_bilingual, Z_ad_se	0.106	0.106	1	0.1055	1
table_c10	bush_bilingual, Z_survey	-0.179	-0.179	1	-0.1793	1
table_c10	bush_bilingual, Z_survey_se	0.105	0.105	1	0.1055	1
table_c10	bush_bilingual, aic	2144	2144	1	2144	1
table_c10	bush_bilingual, constant	1.02	1.02	1	1.02	1
table_c10	bush_bilingual, constant_se	0.091	0.091	1	0.09133	1
table_c10	bush_bilingual, log_likelihood	-1069	-1069	1	-1069	1
table_c10	coffman_bilingual, N	1680	1680	1	1680	1
table_c10	coffman_bilingual, Z_ad	0.18	0.18	1	0.1803	1
table_c10	coffman_bilingual, Z_ad_se	0.112	0.112	1	0.1118	1
table_c10	coffman_bilingual, Z_survey	-0.233	-0.233	1	-0.2331	1
table_c10	coffman_bilingual, Z_survey_se	0.111	0.111	1	0.1114	1
table_c10	coffman_bilingual, aic	1927	1927	1	1927	1
table_c10	coffman_bilingual, constant	1.072	1.072	1	1.072	1
table_c10	coffman_bilingual, constant_se	0.094	0.094	1	0.0943	1
table_c10	coffman_bilingual, log_likelihood	-960.5	-960.5	1	-960.5	1
table_c10	coffman_monolingual, N	1337	1337	1	1337	1
table_c10	coffman_monolingual, Z_ad	-0.82	-0.82	1	-0.8203	1
table_c10	coffman_monolingual, Z_ad_se	0.128	0.128	1	0.1283	1
table_c10	coffman_monolingual, aic	1489	1489	1	1489	1
table_c10	coffman_monolingual, constant	1.483	1.483	1	1.483	1
table_c10	coffman_monolingual, constant_se	0.097	0.097	1	0.09678	1
table_c10	coffman_monolingual, log_likelihood	-742.4	-742.4	1	-742.4	1
table_c10	vela_bilingual, N	1680	1680	1	1680	1
table_c10	vela_bilingual, Z_ad	0.148	0.148	1	0.1485	1
table_c10	vela_bilingual, Z_ad_se	0.142	0.142	1	0.142	1
table_c10	vela_bilingual, Z_survey	-0.201	-0.201	1	-0.2015	1
table_c10	vela_bilingual, Z_survey_se	0.142	0.142	1	0.1417	1
table_c10	vela_bilingual, aic	1352	1352	1	1352	1
table_c10	vela_bilingual, constant	1.86	1.86	1	1.86	1
table_c10	vela_bilingual, constant_se	0.122	0.122	1	0.1219	1
table_c10	vela_bilingual, log_likelihood	-672.9	-672.9	1	-672.9	1
table_c10	vela_monolingual, N	1336	1336	1	1336	1
table_c10	vela_monolingual, Z_ad	-0.743	-0.743	1	-0.7427	1
table_c10	vela_monolingual, Z_ad_se	0.123	0.123	1	0.123	1
table_c10	vela_monolingual, aic	1587	1587	1	1587	1
table_c10	vela_monolingual, constant	1.268	1.268	1	1.268	1
table_c10	vela_monolingual, constant_se	0.093	0.093	1	0.09305	1
table_c10	vela_monolingual, log_likelihood	-791.6	-791.6	1	-791.6	1
text	appendixC_ame_vs_ols, max_abs_difference	0.005			1.571e-06	1
text	p13_general_bush_bilingual, estimate	4.9			4.897	1
text	p13_general_bush_bilingual, std_error	2.3			2.322	1
text	p13_general_coffman_bilingual, estimate				0.2706	
text	p13_general_coffman_bilingual, std_error				2.362	
text	p13_general_coffman_monolingual, estimate	-18.7			-18.72	1
text	p13_general_coffman_monolingual, std_error	2.6			2.645	1
text	p13_general_vela_bilingual, estimate	4.9			4.855	1
text	p13_general_vela_bilingual, std_error	2.4			2.39	1
text	p13_general_vela_monolingual, estimate				-2.012	
text	p13_general_vela_monolingual, std_error				2.614	
text	p13_survey_language_vela_bilingual, estimate	7.3			7.296	1
text	p13_survey_language_vela_bilingual, std_error	2.4			2.39	1
text	p14_cares_coffman_monolingual, estimate	-15.5			-15.52	1
text	p14_cares_coffman_monolingual, std_error	2.4			2.389	1
text	p14_cares_vela_monolingual, estimate	-15.2			-15.2	1
text	p14_cares_vela_monolingual, std_error	2.5			2.466	1
text	p16_coffman_monolingual_democrat, estimate	-14.1			-14.05	1
text	p16_coffman_monolingual_democrat, std_error	3.3			3.34	1
text	p16_coffman_monolingual_republican, estimate	-16.8			-16.78	1
text	p16_coffman_monolingual_republican, std_error	4			3.999	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. (*continued*)

Location	Claim	Paper	Archive	Match	Rewrite	Match
text	p17_linked_fate_bush_bilingual, estimate	0.24			0.2427	1
text	p17_linked_fate_bush_bilingual, std_error	0.04			0.0403	1
text	p17_linked_fate_vela_coffman_bilingual, estimate	0.13			0.1311	1
text	p17_linked_fate_vela_coffman_bilingual, std_error	0.04			0.04127	1

5 Table 2 has no code

Table 2 compares the bilingual Lucid sample in the Bush experiment against bilingual respondents in the 2006 Latino National Survey and the 2012 Pew National Survey of Latinos. Its Lucid column is nine numbers computed from `study_1.csv`, and no script in the deposit computes them. The archive’s `FC_bilinguals_in_text_stats.R` ends with a summary of five covariates from `study_3.csv`, which is a different sample and a different set of variables; nothing produces Table 2.

The rewrite adds `table_2_sample_comparison.R`, and all 17 values reproduce. Doing so requires one undocumented recode: `educ_5` stores “Not Found” as 99 for 96 of the 1,862 respondents, and leaving those in returns a mean of 7.95 against the article’s 3.01. The LNS and Pew columns come from surveys that are not part of the deposit and cannot be checked here.

Table 6: Table 2, Lucid column: the article’s value against the rewrite, which is the only code that computes it.

Quantity	Paper	Rewrite mean	Rewrite SE	Non-missing
female	0.70	0.70	0.01	1862
age	34.80	34.80	0.30	1862
education_5	3.01	3.01	0.03	1766
mexican	0.49	0.49	0.01	1862
cuban	0.07	0.07	0.01	1862
other_hispanic	0.44	0.44	0.01	1862
income_7	4.11	4.11	0.05	1713
income_9	4.49	4.49	0.05	1713
n	1862.00	1862.00	NA	1862

6 Maintained rewrite

The rewrite lives in `maintained/`: a shared `helpers.R` and fourteen scripts covering the seven main-text tables, the ten appendix tables, the three figures and the numbers stated in prose. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

6.1 Architecture

`helpers.R` loads the three deposited CSVs and builds the three analysis subsamples, which is the one piece of data preparation the archive repeats inside each of its eight scripts. Each table script then declares its models as a list of specifications and fits them in one pass, so the outcome, the treatment indicator and the sample of every column are visible in a single block.

Standard errors are the main substitution. The archive fits with `lm()` and passes `estimatr::starprep()` into `stargazer` to get HC2 standard errors, which means the coefficients and the standard errors printed beside them come from two different objects. The rewrite fits with `lm_robust(se_type = "HC2")`, which is the same estimator and the same standard errors in one call, and writes a tidy CSV rather than a LaTeX table, so the table and any downstream check cannot disagree.

`text_in_text_claims.R` reads the committed table output and reports the numbers the article states in prose. It reads no data of its own, so an in-text claim and the table it is drawn from cannot disagree.

6.2 Deprecated patterns replaced

Table 7: Patterns replaced in the maintained rewrite, and whether the replacement was forced or chosen.

Original pattern	Replacement	Required?
<code>'rm(list = ls())'</code>	(omitted)	no
commented <code>'setwd()'</code> instruction	<code>'here::here()'</code>	no
<code>'lm()'</code> + <code>'starprep()'</code> into <code>'stargazer'</code>	<code>'estimatr::lm_robust(se_type = "HC2")'</code>	no
<code>'stargazer'</code> LaTeX to the console	<code>'write_csv()'</code> to <code>'output/'</code>	no
<code>'geom_errorbarh()'</code>	<code>'geom_linerange()'</code>	yes, deprecated in ggplot2 4.0.0
<code>'ggstance::position_dodgev()'</code>	<code>'position_dodge()'</code>	no, ggstance still installs
<code>'margins::margins()'</code>	<code>'marginaleffects::avg_slopes()'</code>	no, margins 0.3.28 released July 2024
<code>'ggsave()'</code> to a bare relative path	<code>'ggsave(here::here("maintained", "output", ...))'</code>	yes, it overwrites the deposit
<code>'expand_grid()'</code> + <code>'for'</code> loop over rows	<code>'expand_grid()'</code> + <code>'mutate()'</code>	no
<code>'within()'</code> blocks over base indexing	<code>'mutate()'</code> paragraphs	no
magrittr pipe	native pipe	no

Only two of the eleven substitutions were forced. `geom_errorbarh()` is deprecated and the horizontal intervals now come from `geom_linerange()`, which the project's style guide prefers in any case. The `ggsave()` paths had to move because the originals write into the deposit. Everything else is a choice, and the table says so rather than implying a package failure that did not occur. `margins` in particular is alive, and running it beside `avg_slopes()` on the same

fitted models returns the same average marginal effects to within $1e-4$, the gap being numerical differentiation in one and not the other.

7 Figures

Figure 1 and Figure 2 are the two the deposit pre-renders, and the rewrite reproduces both. The estimates behind every point are written to `figure_1_main_effects.csv` and `figure_2_het_fx_party.csv`, so the figures are diffable rather than only viewable. Neither deposited figure script prints or saves a single number, so those two CSVs are the only form in which the plotted estimates exist anywhere; `build_ground_truth.R` compares all 52 of them, on estimate, standard error and both confidence limits, against the models the deposited scripts specify.

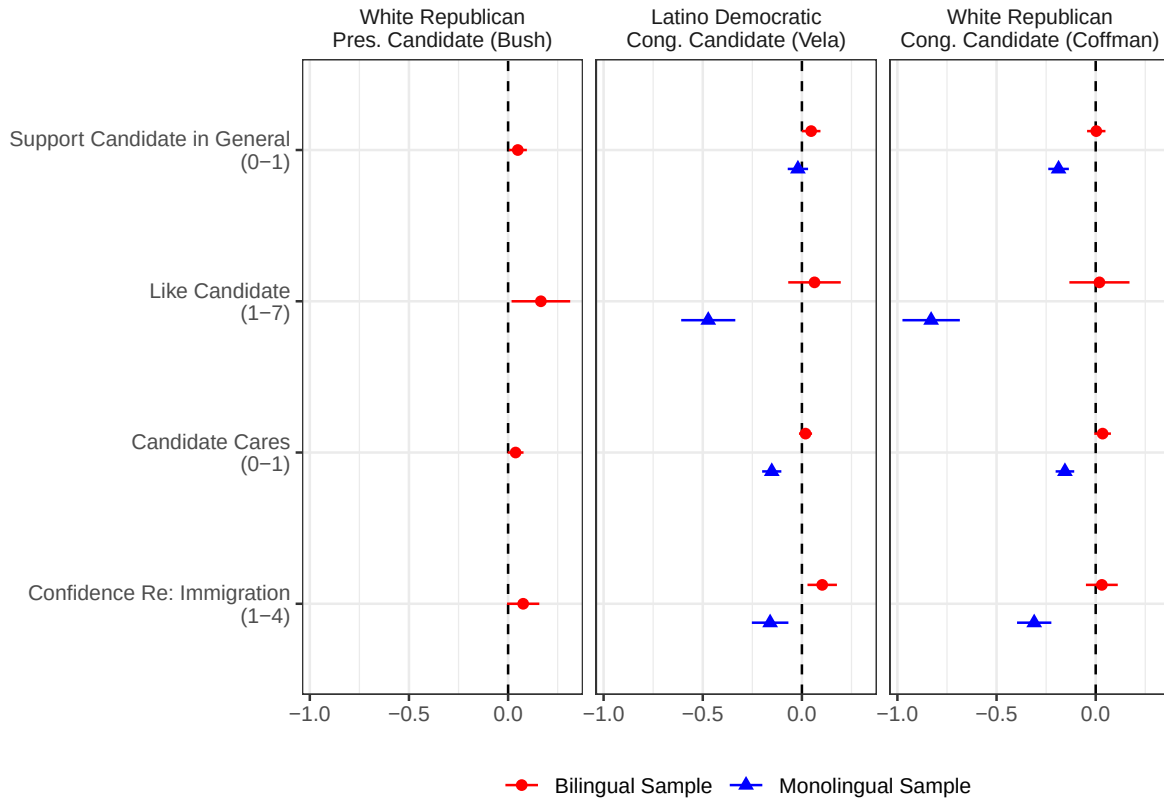


Figure 1: Figure 1 as reproduced by the maintained rewrite.

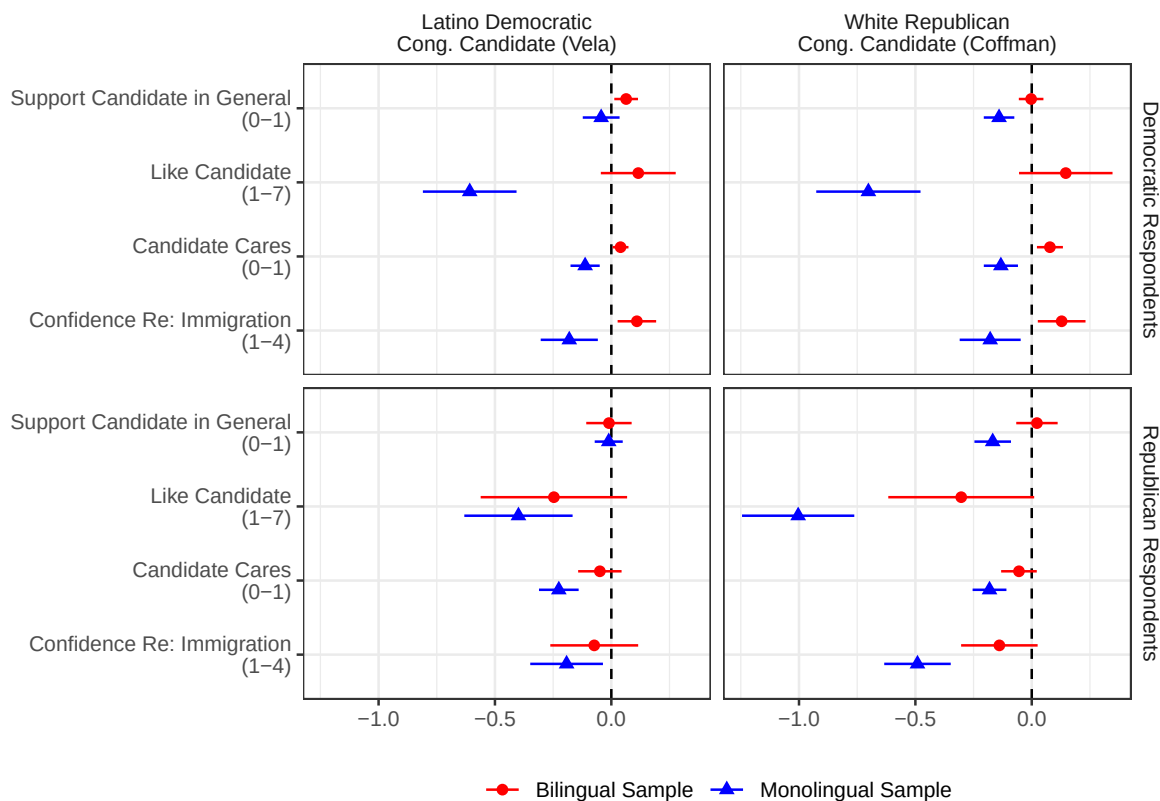


Figure 2: Figure 2 as reproduced by the maintained rewrite.

Figure 3 is the exception to the rule that every number in `output/` comes from data, and it is worth being precise about why. The figure is not an estimate but a calibration exercise: it maps the net electoral payoff of a Spanish-language strategy over two quantities the experiments do not measure, the share of bilinguals in the electorate and the probability that a monolingual sees the ad anyway. The two effect sizes it holds fixed, positive 5 points among bilinguals and negative 15 among monolinguals, are stated in the article's own prose as a supposition. Substituting the estimated effects would draw a different figure than the paper drew, which is a reanalysis rather than a maintenance fix, so the assumption stays and `text_in_text_claims.R` reports the estimates it approximates alongside it. `figure_3_simulation.csv` records the surface on a coarse grid and the break-even mistargeting risk at each bilingual share, which is what the figure is read for.

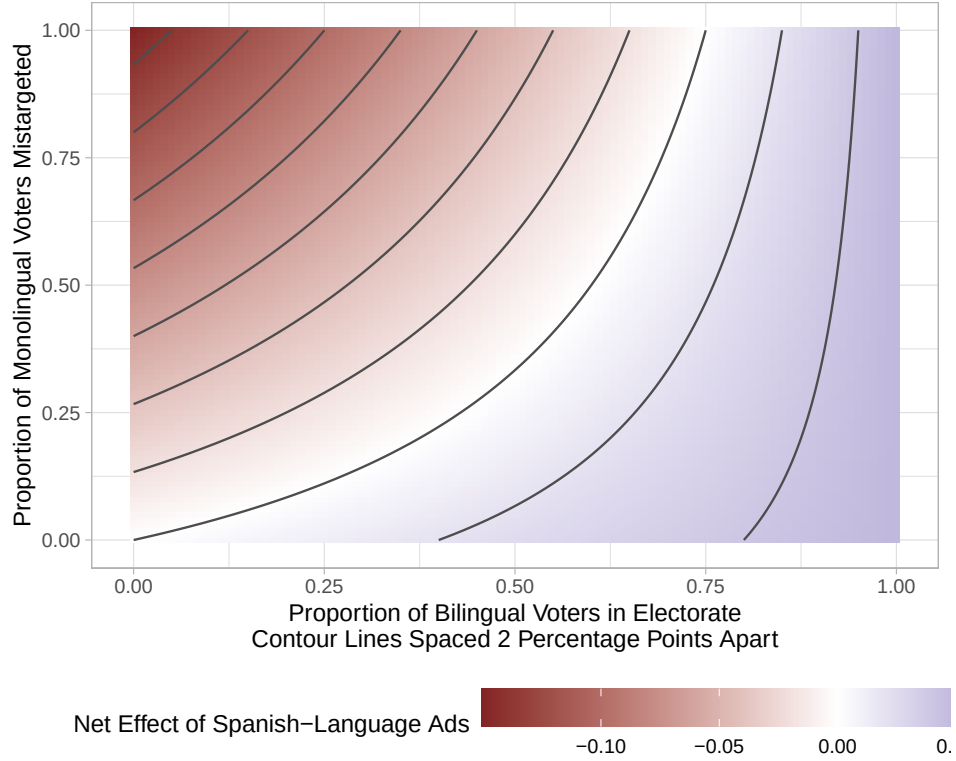


Figure 3: Figure 3 as reproduced by the maintained rewrite.

8 In-text claims

The article states 20 numbers in prose rather than in a table, plus one claim about how closely two estimators agree. All of them reproduce. A further 4 quantities the rewrite computes are the treatment effects the article discusses without printing, so they carry no comparison target and are recorded with a blank paper value rather than dropped.

Table 8: Numbers the article states in prose, against the maintained rewrite.

Claim	Paper	Rewrite	Match
appendixC_ame_vs_ols, max_abs_difference	0.005	1.571e-06	1
p13_general_bush_bilingual, estimate	4.9	4.897	1
p13_general_bush_bilingual, std_error	2.3	2.322	1
p13_general_coffman_bilingual, estimate		0.2706	
p13_general_coffman_bilingual, std_error		2.362	
p13_general_coffman_monolingual, estimate	-18.7	-18.72	1
p13_general_coffman_monolingual, std_error	2.6	2.645	1
p13_general_vela_bilingual, estimate	4.9	4.855	1
p13_general_vela_bilingual, std_error	2.4	2.39	1
p13_general_vela_monolingual, estimate		-2.012	

Table 8: Numbers the article states in prose, against the maintained rewrite. (*continued*)

Claim	Paper	Rewrite	Match
p13_general_vela_monolingual, std_error		2.614	
p13_survey_language_vela_bilingual, estimate	7.3	7.296	1
p13_survey_language_vela_bilingual, std_error	2.4	2.39	1
p14_cares_coffman_monolingual, estimate	-15.5	-15.52	1
p14_cares_coffman_monolingual, std_error	2.4	2.389	1
p14_cares_vela_monolingual, estimate	-15.2	-15.2	1
p14_cares_vela_monolingual, std_error	2.5	2.466	1
p16_coffman_monolingual_democrat, estimate	-14.1	-14.05	1
p16_coffman_monolingual_democrat, std_error	3.3	3.34	1
p16_coffman_monolingual_republican, estimate	-16.8	-16.78	1
p16_coffman_monolingual_republican, std_error	4	3.999	1
p17_linked_fate_bush_bilingual, estimate	0.24	0.2427	1
p17_linked_fate_bush_bilingual, std_error	0.04	0.0403	1
p17_linked_fate_vela_coffman_bilingual, estimate	0.13	0.1311	1
p17_linked_fate_vela_coffman_bilingual, std_error	0.04	0.04127	1

The first row is the appendix’s claim that the average marginal effects from the logit models “match the OLS models to the second decimal place or better”. Across the ten models the largest gap is 1.6e-06, three orders of magnitude tighter than the claim.

9 Maintained rewrite verification

Every one of the 541 claims with a published value to compare against matches it, so no row carries a `defect_locus`. Two full runs of `run_all.R` produce byte-identical CSV and PNG output and an identical ground truth. Only the PDF figures differ between runs, because a PDF records the time it was written. Nothing in the pipeline draws a random number, so there is no seed to pin and no sampler change to inherit.

10 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-01

Table 10: Package versions used for the run behind this report.

Package	Version
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estimatr	1.0.6
marginaleffects	0.32.0
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2
